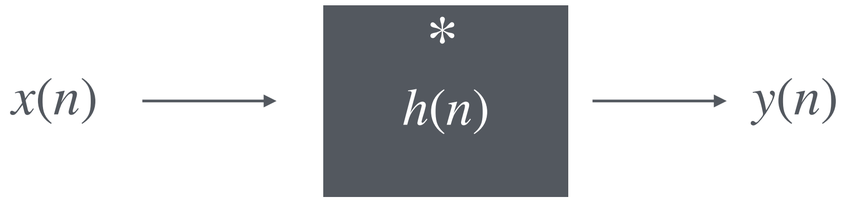
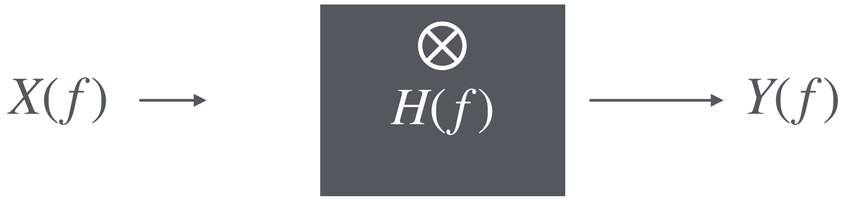
Chapter 5

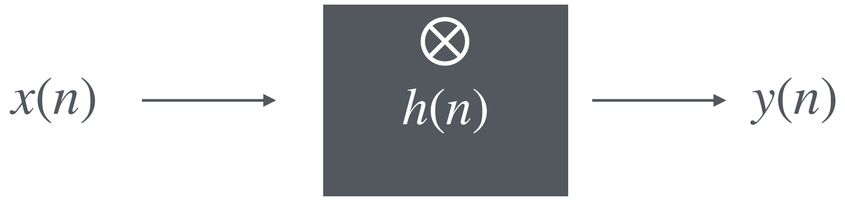
FIR

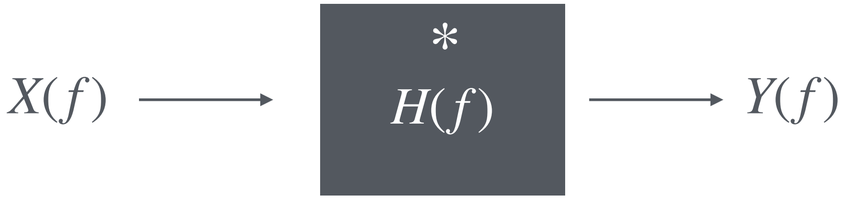
HW





Convolution on one domain =
Multiplication in other domain





FIR Lowpass Filters

- Simple symmetric coefficients
- Chebyshev window
- Kaiser window
- Blackman window

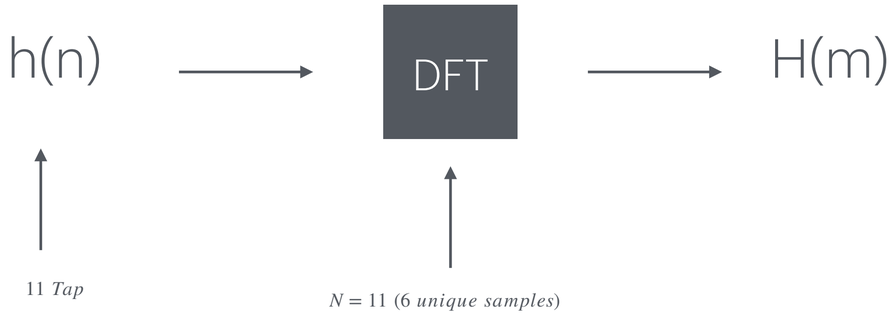
Digital Low Pass Filter Ripples

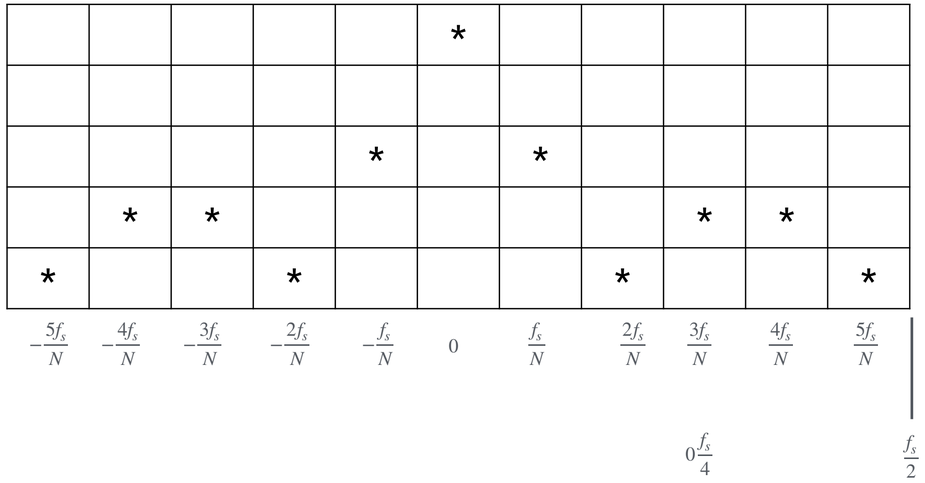
Ripples persists in all fir filters because its not possible to synthesize an infinite time sinc-shape function in time (i.e. low pass filter window in frequency domain)

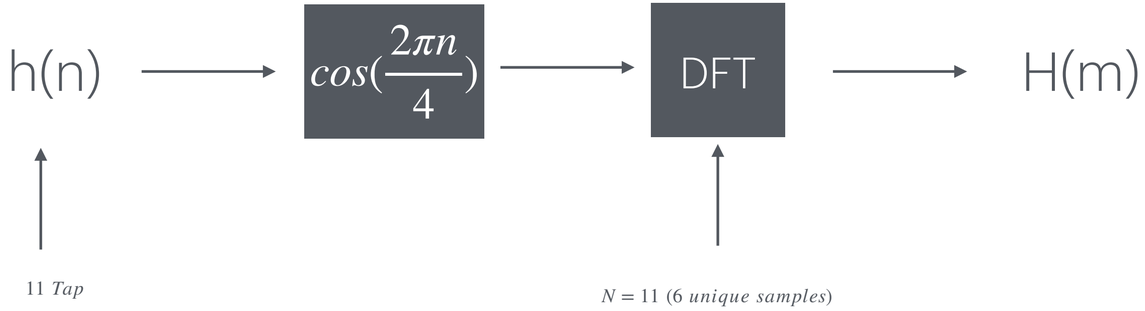
Low Pass Resolution

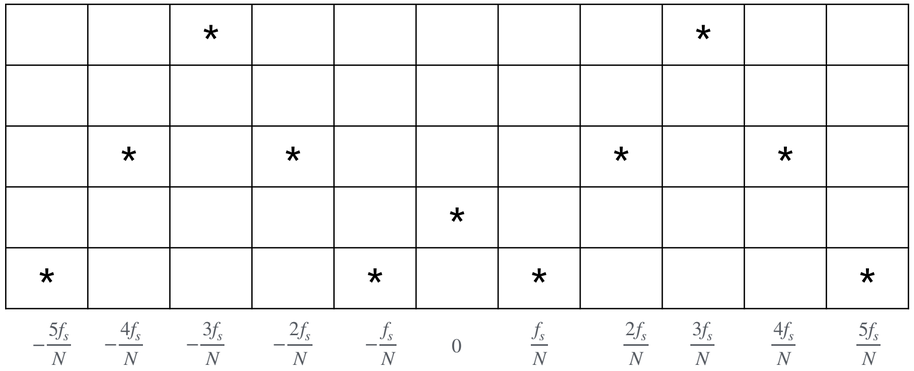
Low pass filters without gradual fall of sinc-shape impulse response will reduce frequency spectrum side lobes with an increase in mainline width.

Bandpass FIR



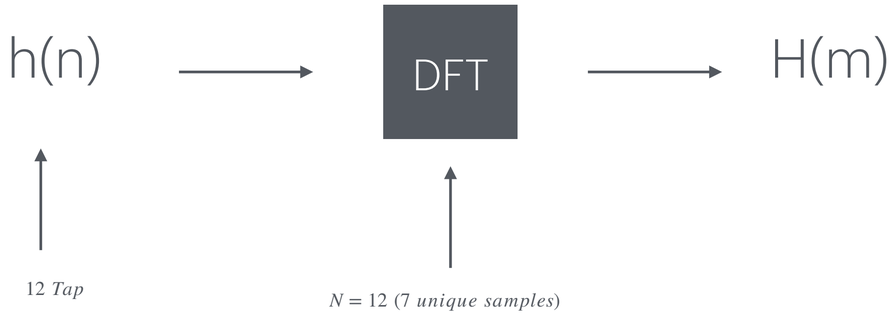


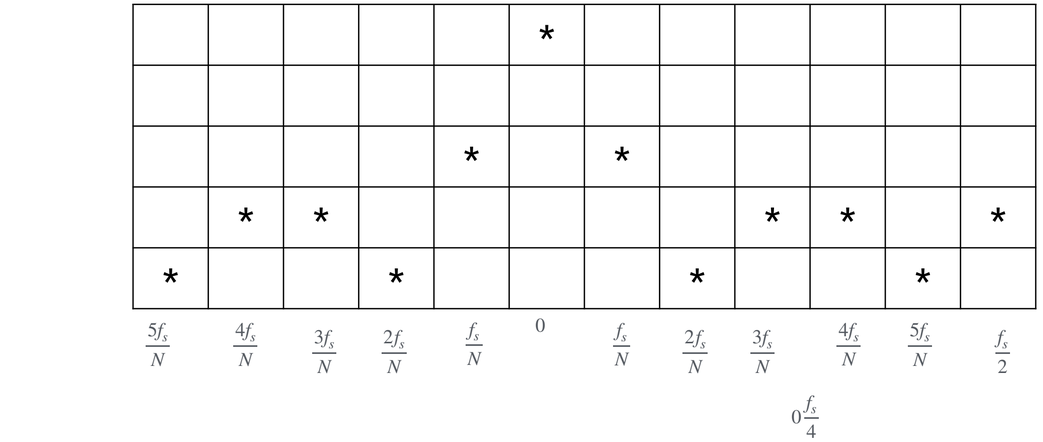


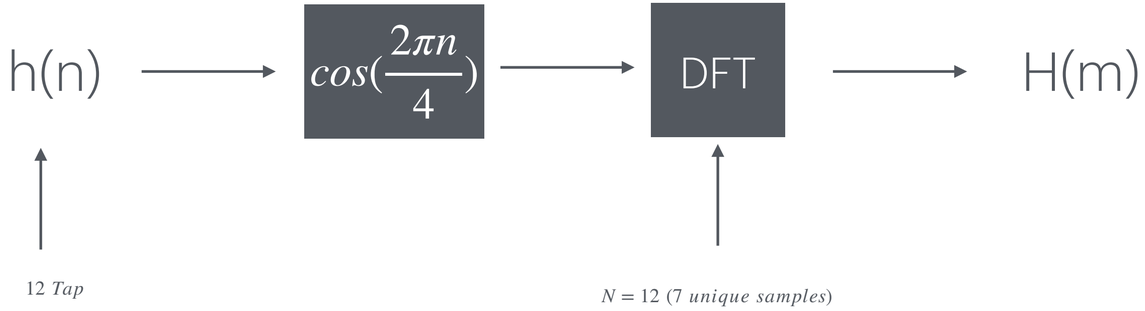


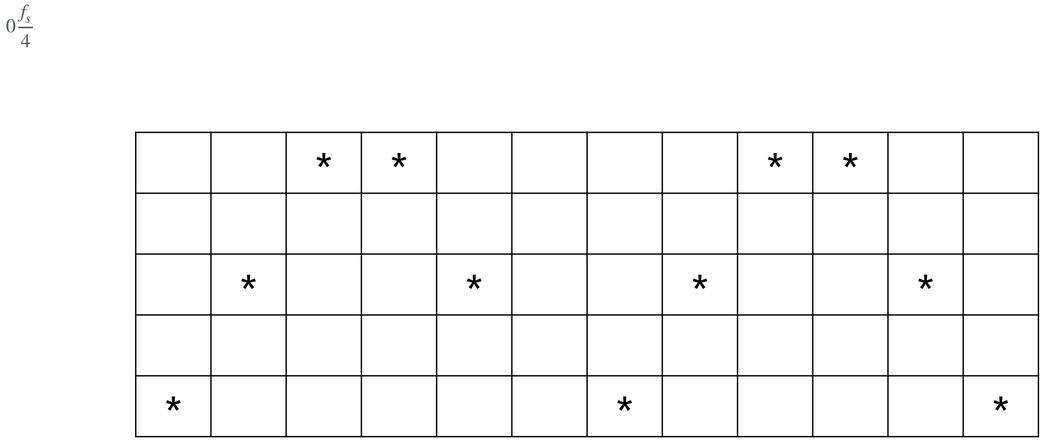
Amplitude halved

Bandpass FIR



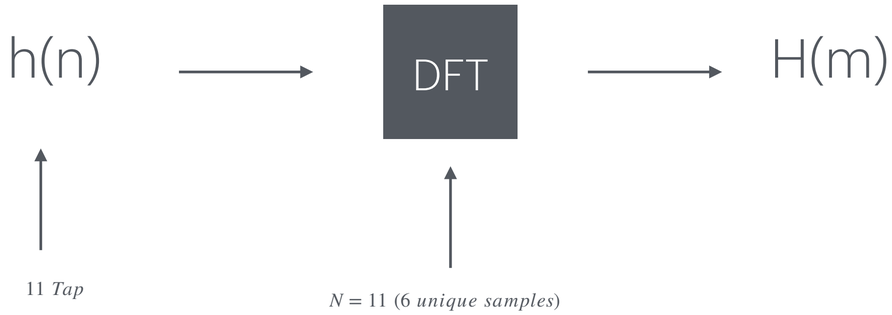


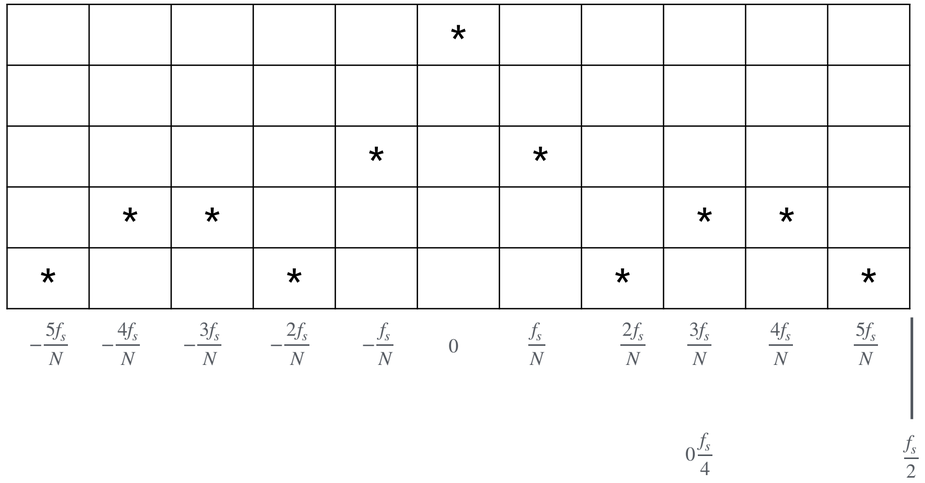


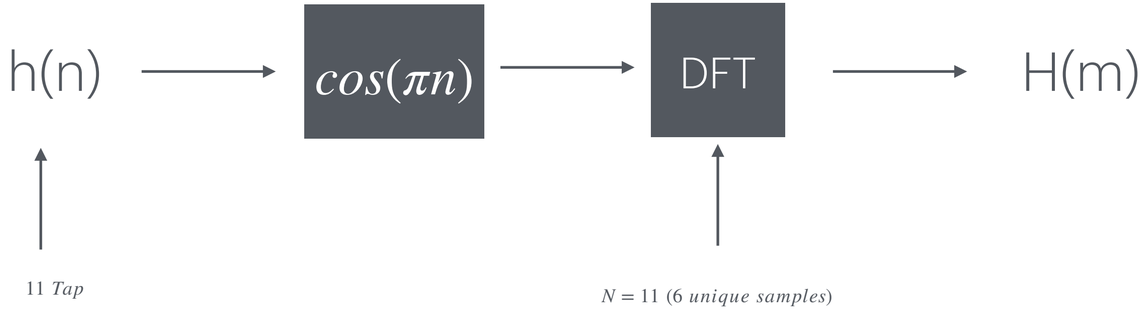


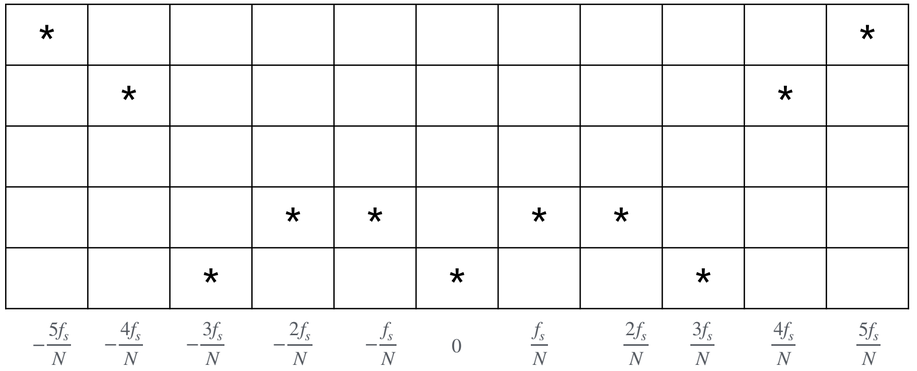
Amplitude halved

Highpass FIR









Amplitude halved

Discrete Convolution

$x(n)$ $h(n)$ = $y(n)$

11 samples 32 samples

42 (11 + 32 - 1) samples (number of operations)

$$\begin{bmatrix} Y_0 \\ Y_1 \\ Y_2 \\ \cdot \\ Y_4 \end{bmatrix} = \begin{bmatrix} X_0 & X_{-1} & X_{-2} & \cdot \\ X_1 & X_0 & X_{-1} & \cdot \\ X_2 & X_1 & X_0 & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ X_{N-1} & X_{N-2} & X_{N-3} & \cdot \end{bmatrix} \bullet \begin{bmatrix} H_0 \\ H_1 \\ H_2 \\ \vdots \\ H_{N-1} \end{bmatrix}$$